

Project Summary: House Price Prediction

This project involved predicting house prices using a dataset of 545 listings. We trained both a Linear Regression model and a Random Forest Regressor.

Key Findings & Model Accuracy:

- The **Linear Regression** model performed slightly better, achieving an R^2 of approximately **0.559** and an average prediction error (MAE) of about **₹1.11 million**. This means the model can explain roughly 56% of the variation in house prices, with predictions typically within ₹1.11 million of the actual price.
- The **Random Forest** model was similar, with an R^2 of around **0.500** and an MAE of **₹1.12 million**.

Which features influence house price the most?

- **Area** is by far the most influential feature, accounting for over 55% of the importance in the Random Forest model. Other significant factors include **bathrooms**, **parking**, **stories**, and the **furnishing status** (especially 'unfurnished' having a notable negative impact).

What surprised you in the data?

- A surprising observation was that features such as `mainroad`, `guestroom`, `basement`, `hotwaterheating`, `airconditioning`, and `prefarea` showed **zero impact** on price in both models. This suggests these binary categorical columns either had uniform values in the dataset (zero variance), were perfectly correlated with other features, or were affected by an issue in the data cleaning process that rendered them constant.

One recommendation for a real estate business:

- Prioritize highlighting property **area**, the number of **bathrooms**, **parking availability**, and the number of **stories** in listings, as these are the most impactful features for predicting price. Furthermore, closely examine the binary amenities (`airconditioning`, `mainroad`, `prefarea`, etc.) for their true influence on pricing by ensuring sufficient variance in the data or by collecting richer data for these aspects.